Q1 - 25 July - Shift 1

If the numbers appeared on the two throws of a fair six faced die are α and β, then the probability that $x^2 + \alpha x + \beta > 0$, for all $x \in R$, is :

(A) $\frac{17}{36}$ (B) $\frac{4}{9}$ (C) $\frac{1}{2}$ (D) $\frac{19}{36}$

Space for your notes:

Q2 - 25 July - Shift 1

Let a, b be two non-zero real numbers. If p and r are the roots of the equation $x^2 - 8ax + 2a = 0$ and q and s are the roots of the equation $x^2 + 12bx + 6b = 0$, such that $\frac{1}{p}, \frac{1}{q}, \frac{1}{r}, \frac{1}{s}$ are in A.P., then $a^{-1} - b^{-1}$ is equal to _______.

Space for your notes:

Q3 - 26 July - Shift 1

If for some p, q, r $\in$ R, not all have same sign, one of the roots of the equation $(p^2 + q^2)x^2 - 2q(p + r)x + q^2 + r^2 = 0$ is also a root of the equation $x^2 + 2x - 8 = 0$, then $\frac{q^2 + r^2}{p^2}$ is equal to-

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Q4 - 26 July - Shift 1

The number of distinct real roots of the equation $x^5(x^3 - x^2 - x + 1) + x(3x^3 - 4x^2 - 2x + 4) - 1 = 0$ is

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Q5 - 26 July - Shift 2

The minimum value of the sum of the squares of the roots of $x^2+(3-a)x+1=2a$ is:

(A) 4 (B) 5

(C) 6 (D) 8

Space for your notes:

Q6 - 26 July - Shift 2

Let the abscissae of the two points P and Q on a circle be the roots of $x^2 - 4x - 6 = 0$ and the ordinates of P and Q be the roots of $y^2 + 2y - 7 = 0$. If PQ is a diameter of the circle $x^2 + y^2 + 2ax + 2by + c = 0$, then the value of $(a+b-c)$ is

(A) 12 (B) 13 (C) 14 (D) 16

Space for your notes:

Q7 - 27 July - Shift 2

If α, β are the roots of the equation

$$x^2 - \left(5 + 3^{\sqrt{\log_3 5}} - 5^{\sqrt{\log_5 3}}\right) + 3\left(3^{(\log_3 5)^{\frac{1}{3}}} - 5^{(\log_5 3)^{\frac{2}{3}}} - 1\right) = 0$$

then the equation, whose roots are

$\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$,

(A) $3x^2 - 20x - 12 = 0$

(B) $3x^2 - 10x - 4 = 0$

(C) $3x^2 - 10x + 2 = 0$

(D) $3x^2 - 20x + 16 = 0$

Space for your notes:

Q8 - 28 July - Shift 1

The sum of all real values of x for which $\frac{3x^2-9x+17}{x^2+3x+10}=\frac{5x^2-7x+19}{3x^2+5x+12}$ is equal to ____.

Space for your notes:

Q9 - 28 July - Shift 2

Let α, β be the roots of the equation $x^2-\sqrt{2}x+\sqrt{6}=0$ and $\frac{1}{\alpha^2}+1, \frac{1}{\beta^2}+1$ be the roots of the equation $x^2+ax+b=0$. Then the roots of the equation $x^2-(a+b-2)x+(a+b+2)=0$ are :

(A) non-real complex numbers

(B) real and both negative

(C) real and both positive

(D) real and exactly one of them is positive

Space for your notes:

Q10 - 28 July - Shift 2

Let $f(x)=ax^2+bx+c$ be such that $f(1)=3$, $f(-2)=\lambda$ and $f(3)=4$. If $f(0)+f(1)+f(-2)+f(3)=14$, then λ is equal to

(A) -4

(B) $\frac{13}{2}$

(C) $\frac{23}{2}$

(D) 4

Space for your notes:

Q11 - 29 July - Shift 2

Let α, β (α > β) be the roots of the quadratic equation $x^2-x-4=0$. If $P_n=\alpha^n-\beta^n, n\in\mathbb{N}$, then $\frac{P_{15}P_{16}-P_{14}P_{16}-P_{15}^2+P_{14}P_{15}}{P_{13}P_{14}}$ is equal to ______.

Space for your notes:

Answer Key

Q1 (A)	**Q2** (38)	**Q3** (272)	**Q4** (3)
Q5 (C)	**Q6** (A)	**Q7** (B)	**Q8** (6)
Q9 (B)	**Q10** (D)	**Q11** (16)	

Q1 (A)

$x^2 + \alpha x + \beta > 0, \forall x \in R$

$D = \alpha^2 - 4\beta < 0$

$\alpha^2 < 4\beta$

Total cases $= 6 \times 6 = 36$

Fav. cases $= \beta = 1, \alpha = 1$

$\beta = 2, \alpha = 1, 2$

$\beta = 3, \alpha = 1, 2, 3$

$\beta = 4, \alpha = 1, 2, 3$

$\beta = 5, \alpha = 1, 2, 3, 4$

$\beta = 6, \alpha = 1, 2, 3, 4$

Total favourable cases $= 17$

$$P(x) = \frac{17}{36}$$

Q2 (38)

$x^2 - 8ax + 2a = 0$ $\quad$ $x^2 + 12bx + 6b = 0$

$p + r = 8a$ $\quad$ $q + s = -12b$

$pr = 2a$ $\quad$ $qs = 6b$

$\frac{1}{p} + \frac{1}{r} = 4$ $\quad$ $\frac{1}{q} + \frac{1}{s} = -2$

$\frac{2}{q} = 4$ $\quad$ $\frac{2}{r} = -2$

$q = \frac{1}{2}$ $\quad$ $r = -1$

$p = \frac{1}{5}$ $\quad$ $s = \frac{-1}{4}$

Now, $\frac{1}{a} - \frac{1}{b} = \frac{2}{pr} - \frac{6}{qs} = 38$

Q3 (272)

$$(px - q)^2 + (qx - r)^2 = 0$$

$$\Rightarrow x = \frac{q}{p} = \frac{r}{q} = -4$$

$$\Rightarrow \frac{q^2 + r^2}{p^2} = 272$$

Q4 (3)

$x^5(x^3 - x^2 - x + 1) + x(3x^3 - 4x^2 - 2x + 4) - 1 = 0$

$\Rightarrow (x-1)^2 (x + 1) (x^5 + 3x - 1) = 0$

Let $f(x) = x^5 + 3x - 1$

$f'(x) > 0 \ \forall \ x \in R$

Hence 3 real distinct roots.

Q5 (C)

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

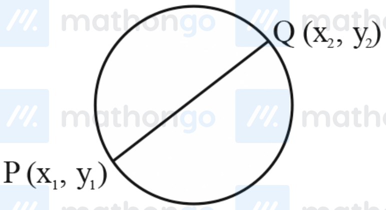
let $f(a) = (3 - a)^2 - 2(1 - 2a)$

$f(a) = a^2 - 2a + 7$

$f(a) = (a - 1)^2 + 6$

$$\left.f(a)\right)_{\min.} = 6$$

Q6 (A)



Equation of circle diameter form

$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

(where x_1 , x_2 are the roots of $x^2 - 4x - 6 = 0$ and y_1, y_2 are the roots of $y^2 + 2y - 7 = 0$)

$x^2 + y^2 - 4x + 2y - 13 = 0$

Now,

Compare it with the given equation, we get

$a = -2, b = 1, c = -13$

Now

$a + b - c = 12$

Q7 (B)

Bonus because 'x' is missing the correct will be,

$$x^2-\left(5+3^{\sqrt{\log_3 5}}-5^{\sqrt{\log_5 3}}\right)x+3\left(3^{(\log_3 5)^{\frac{1}{3}}}-5^{(\log_5 3)^{\frac{2}{3}}}-1\right)=0$$

$$3^{\sqrt{\log_3 5}}=3^{\sqrt{\log_3 5}.\sqrt{\log_3 5}.\sqrt{\log_5 3}}=3^{\log_3 5.\sqrt{\log_5 3}}$$

$$=\left(3^{\log_3 5}\right)^{\sqrt{\log_5 3}}=5^{\sqrt{\log_5 3}}$$

$$3^{\sqrt[3]{\log_3 5}}=3^{\log_3 5.\sqrt[3]{(\log_5 3)^2}}=\left(3^{\log_3 5}\right)^{(\log_5 3)^{2/3}}$$

$$=5^{(\log_5 3)^{2/3}}$$

So, equation is $x^2 - 5x - 3 = 0$ and roots are α & β

$\{\alpha + \beta = 5; \alpha\beta = -3\}$

New roots are $\alpha+\frac{1}{\beta}$ & $\beta+\frac{1}{\alpha}$

i.e., $\frac{\alpha\beta+1}{\beta}$ & $\frac{\alpha\beta+1}{\alpha}$ i.e., $\frac{-2}{\beta}$ & $\frac{-2}{\alpha}$

Let $\frac{-2}{\alpha}=t \Rightarrow \alpha=\frac{-2}{t}$

As $\alpha^2-5\alpha-3=0$

$$\Rightarrow \left(\frac{-2}{t}\right)^2-5\left(\frac{-2}{t}\right)-3=0$$

$$\Rightarrow \frac{4}{t^2}+\frac{10}{t}-3=0$$

$\Rightarrow 4 + 10t - 3t^2 = 0$

$\Rightarrow 3t^2 - 10t - 4 = 0$

i.e., $3x^2 - 10x - 4 = 0$

Q8 (6)

$$\frac{3x^2-9x+17}{x^2+3x+10}=\frac{5x^2-7x+19}{3x^2+5x+12}$$

$$\frac{x^2+3x+10+2x^2-12x+7}{x^2+3x+10}=\frac{3x^2+5x+12+2x^2-12x+7}{3x^2+5x+12}$$

$$1+\frac{2x^2-12x+7}{x^2+3x+10}=1+\frac{2x^2-12x+7}{3x^2+5x+12}$$

$$\left(2x^2-12x+7\right)\left(\frac{1}{x^2+3x+10}-\frac{1}{3x^2+5x+12}\right)=0$$

$$2x^2-12x+7=0 \text{ OR } 3x^2+5x+12=x^2+3x+10$$

$$x=\frac{12\pm\sqrt{D}}{4} \qquad 2x^2+2x+2=0$$

$$x^2+x+1=0$$

Sum of Roots = 6 No solution.

Q9 (B)

$$a = \frac{-1}{\alpha^2} - \frac{1}{\beta^2} - 2$$

$$b = \frac{1}{\alpha^2} + \frac{1}{\beta^2} + 1 + \frac{1}{\alpha^2\beta^2}$$

$$a + b = \frac{1}{(\alpha\beta)^2} - 1 = \frac{1}{6} - 1 = -\frac{5}{6}$$

$$x^2 - \left(-\frac{5}{6} - 2\right)x + \left(2 - \frac{5}{6}\right) = 0$$

$$6x^2 + 17x + 7 = 0$$

$x = -\frac{7}{3}$, $x = -\frac{1}{2}$ are the roots

Both roots are real and negative.

Q10 (D)

$$f(0) + 3 + \lambda + 4 = 14$$

$$\therefore \quad f(0) = 7 - \lambda = c$$

$$f(1) = a + b + c = 3 \qquad \ldots(i)$$

$$f(3) = 9a + 3b + c = 4 \qquad \ldots(ii)$$

$$f(-2) = 4a - 2b + c = \lambda \qquad \ldots(iii)$$

(ii) – (iii)

$a + b = \dfrac{4-\lambda}{5}$ put in equation (i)

$$\frac{4-\lambda}{5} + 7 - \lambda = 3$$

$$6\lambda = 24; \quad \lambda = 4$$

Q11 (16)

$Pn = \alpha^n - \beta^n$ $x^2 - x - 4 = 0$

$$\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}} \text{\_\_(1)}$$

As $P_n - P_{n-1} = (\alpha^n - \beta^n) - (\alpha^{n-1} - \beta^{n-1})$

$= \alpha^{n-2}(\alpha^2 - \alpha) - \beta^{n-2}(\beta^2 - \beta)$

$= 4(\alpha^{n-2} - \beta^{n-2})$

$P_n - P_{n-1} = 4P_{n-2}$

Hence Expression (1)

$$\frac{P_{16}(P_{15} - P_{14}) - P_{15}(P_{15} - P_{14})}{P_{13}P_{14}}$$

$$= \frac{(P_{15} - P_{14})(P_{16} - P_{15})}{P_{13}P_{14}} = \frac{(4P_{13})(4P_{14})}{P_{13}P_{14}} = 16$$